Physics-Informed LiquidGAN for Zero-Shot Cross-Domain Thermal Image Synthesis

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Introduction – Problem Context

- Thermal imaging used in healthcare, surveillance, agriculture.
- Deep learning requires large annotated datasets.
- Thermal datasets are limited and expensive to collect.
- Data scarcity reduces model generalization.

Need: Physically consistent data augmentation.

Introduction – Key Challenge

- GANs used for data augmentation [1].
- Conventional GANs treat thermal images as grayscale.
- Ignore heat diffusion physics:

$$\nabla^2 T = 0$$

Problem: Unrealistic thermal gradients and poor cross-domain robustness.

Literature Review (1/5): Generative Adversarial Networks

Paper: Goodfellow et al., "Generative Adversarial Networks" [1]
Methodology:

- Introduced Generator–Discriminator adversarial framework.
- Learns data distribution via minimax optimization.
- Widely adopted for image synthesis and augmentation.

Limitations:

- Requires large datasets for stable training.
- No inductive bias for physical processes.
- Purely data-driven — vulnerable to domain shift.

Implication: Motivates structured constraints and robustness mechanisms for scarce thermal data.

Literature Review (2/5): GAN-Based Thermal Synthesis

Paper: Zhang et al., "Infrared Image Generation via GANs" [2]
Methodology:

- CNN-based GAN for infrared/thermal image generation.
- Uses adversarial and pixel-level reconstruction losses.

Limitations:

- Treats thermal data as grayscale intensity.
- Does not enforce heat diffusion or energy conservation laws.
- Limited evaluation under cross-dataset domain shift.

Implication: Thermal synthesis requires thermodynamic constraints for realism and generalization.

Literature Review (3/5): Neural Ordinary Differential Equations

Paper: Chen et al., "Neural Ordinary Differential Equations" [3]
Methodology:

- Models hidden representations using continuous-time dynamics.
- Solves $\frac{dh}{dt} = f(h, t, \theta)$ via numerical ODE solvers.

Limitations:

- Ensures smooth evolution but remains data-driven.
- Does not incorporate governing physical PDEs.
- No explicit robustness to unseen domains.

Implication: Continuous modeling suits thermal dynamics but needs physics-aware regularization.

Literature Review (4/5): Liquid Neural Networks

Paper: Hasani et al., "Liquid Time-Constant Networks" [4]
Methodology:

- Dynamical neural architecture with adaptive internal state evolution.
- Governed by differential equations with learnable time constants.

Limitations:

- Primarily evaluated in control and time-series tasks.
- No thermodynamic priors in visual domains.
- Limited study in adversarial image synthesis.

Implication: Dynamic discriminators improve adaptability but lack physics grounding.

Literature Review (5/5): LiquidGAN

Paper: Sarath & Nair, "LiquidGAN" [5]
Methodology:

- Neural ODE-based generator for smooth latent evolution.
- Liquid Neural Network discriminator for dynamic refinement.
- Designed for thermal image synthesis under data scarcity.

Limitations:

- Fully data-driven; no PDE-based thermal constraints.
- Thermal consistency evaluated only via image metrics (FID, SSIM).
- No zero-shot cross-domain validation.

Implication: Motivates Physics-Constrained and Zero-Shot extension of LiquidGAN.

Research Gap Summary

Area	Key Gap
Physics Modeling	No explicit enforcement of thermodynamic laws (e.g., heat diffusion); visual realism does not guarantee physical consistency.
Physics Validation	Evaluation relies only on image metrics (FID, SSIM, PSNR); lacks PDE-based residual or energy consistency validation.
Zero-Shot Generalization	Training and testing performed within same dataset; no unseen-domain or cross-dataset evaluation.
Cross-Domain Robustness	No mechanism ensuring stable latent behavior under domain shift (sensor variation, environment changes).

Proposed Methodology – Overview

Physics-Constrained LiquidGAN (PC-LiquidGAN)

- Neural ODE-based Generator
- Liquid Neural Network Discriminator
- Physics-aware thermal loss

Generator Objective:

$$L_G^{PC} = L_{adv} + \lambda L_{recon} + \mu_1 L_{flux} + \mu_2 L_{energy}$$

Proposed Methodology – Physics-Constrained LiquidGAN (PC-LiquidGAN) I

1. Neural ODE-based Generator

Latent dynamics are modeled as a continuous-time system:

$$\frac{dz(t)}{dt} = f_\theta(z(t)), \qquad T(t) = G(z(t))$$

- $z(t)$: continuous latent representation
- f_θ : learnable ODE dynamics
- $G(\cdot)$: decoder producing thermal field

This formulation allows direct integration of differential physical laws.

2. Liquid Neural Network Discriminator

$$\frac{dh(t)}{dt} = \frac{1}{\tau}\left(f_\phi(h(t) + x_{enc}) - h(t)\right)$$

Proposed Methodology – Physics-Constrained LiquidGAN (PC-LiquidGAN) II

- $h(t)$: liquid hidden state
- τ : adaptive time constant
- Provides dynamic feature refinement

3. Governing Thermal Physics

Heat diffusion equation:

$$\frac{\partial T(x,t)}{\partial t} = \alpha \nabla^2 T(x,t)$$

Rewriting as residual:

$$\mathcal{R}(x,t) = \frac{\partial T}{\partial t} - \alpha \nabla^2 T$$

For physically valid thermal synthesis:

Proposed Methodology – Physics-Constrained LiquidGAN (PC-LiquidGAN) III

$$\mathcal{R}(x, t) = 0$$

4. Conversion to Trainable Flux Loss

Since exact equality cannot be enforced, we minimize the residual:

$$L_{flux} = \frac{1}{N} \sum_{x} \left(\frac{\partial T}{\partial t} - \alpha \nabla^2 T \right)^2$$

Practical Discretization:

$$\frac{\partial T}{\partial t} \approx \frac{T_{t+\Delta t} - T_{t-\Delta t}}{2\Delta t}$$

$$\nabla^2 T_{i,j} = T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1} - 4T_{i,j}$$

Thus, the physics constraint becomes directly computable during training.

Proposed Methodology – Physics-Constrained LiquidGAN (PC-LiquidGAN) IV

5. Energy Conservation Constraint

Total thermal energy:

$$E(t) = \sum_{i,j} T_{i,j}(t)$$

Energy regularization:

$$L_{energy} = (E(t + \Delta t) - E(t))^2$$

This prevents artificial energy amplification in generated images.

6. Final Physics-Constrained Generator Objective

$$L_G^{PC} = L_{adv} + \lambda L_{recon} + \mu_1 L_{flux} + \mu_2 L_{energy}$$

The generator parameters θ are optimized by minimizing L_G^{PC}, where adversarial realism, reconstruction fidelity, and thermodynamic consistency are jointly enforced.

Proposed Methodology – Implementation I

1. **Terminology Mapping (Physics $\leftrightarrow$ GAN)**
   - $z(t)$: latent feature representation evolving inside the Neural ODE generator
   - t : continuous ODE depth variable (not real physical time)
   - $T(t)$: generated thermal image tensor at ODE state t
   - Δt : numerical step interval used by ODE solver
   - $\nabla^2 T$: spatial diffusion of generated image computed via convolution
   - $\mathcal{R}$: residual measuring deviation from heat diffusion law
   - α : thermal diffusivity parameter (fixed or learnable)

Proposed Methodology – Implementation II

Thus, temporal evolution corresponds to latent-space dynamics within the generator.

2. ODE-Based Temporal Sampling
The Neural ODE generator is solved using a numerical solver (e.g., RK4) to obtain multiple latent states:

$$z_{t-\Delta t}, \quad z_t, \quad z_{t+\Delta t}$$

These are decoded to thermal images:

$$T_{t-\Delta t}, \quad T_t, \quad T_{t+\Delta t}$$

This enables approximation of temporal derivatives required for physics constraints.

3. Temporal Derivative Approximation
The time derivative is computed using central finite difference:

Proposed Methodology – Implementation III

$$\frac{\partial T}{\partial t} \approx \frac{T_{t+\Delta t} - T_{t-\Delta t}}{2\Delta t}$$

This is implemented using tensor subtraction and scaling, ensuring full differentiability during backpropagation.

4. Spatial Laplacian Computation

The Laplacian operator is implemented using a fixed convolution kernel:

$$\nabla^2 T = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} * T$$

Convolution enables efficient and differentiable spatial diffusion modeling.

5. Physics Residual Construction

The residual is computed as:

Proposed Methodology – Implementation IV

$$\mathcal{R} = \frac{\partial T}{\partial t} - \alpha \nabla^2 T$$

The flux loss is defined as:

$$L_{flux} = \|\mathcal{R}\|_2^2$$

This penalizes violations of the heat diffusion equation.

6. Energy Regularization

Total thermal energy is computed by spatial summation:

$$E(t) = \sum_{i,j} T_{i,j}(t)$$

Energy loss:

$$L_{energy} = \left(E(t + \Delta t) - E(t)\right)^2$$

Proposed Methodology – Implementation V

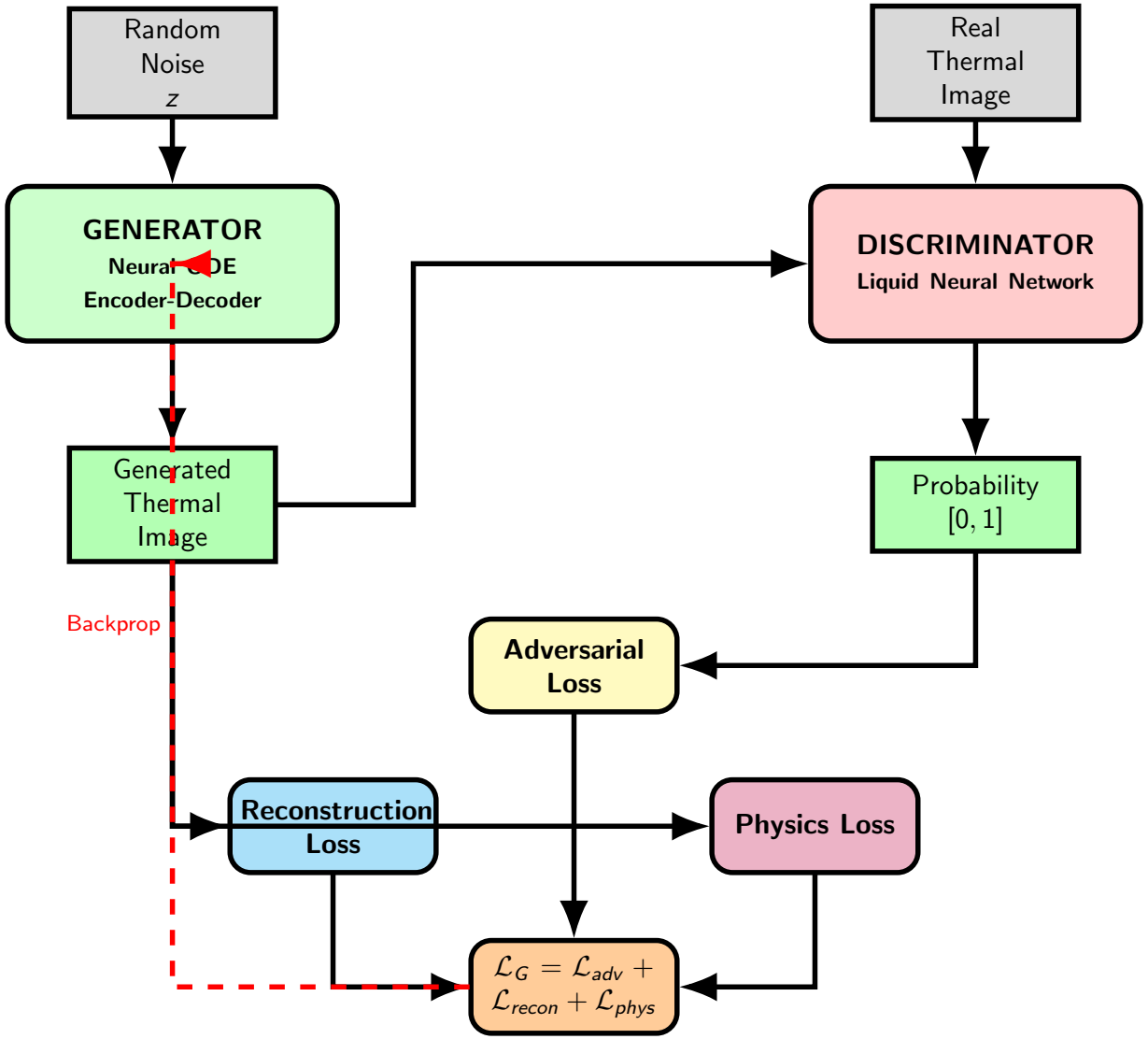
This prevents artificial energy amplification in generated outputs.

7. Integration into GAN Training
The generator is optimized by minimizing:

$$L_G^{PC} = L_{adv} + \lambda L_{recon} + \mu_1 L_{flux} + \mu_2 L_{energy}$$

All physics operations are implemented as differentiable tensor computations, allowing gradients to propagate through the decoder, ODE solver, and generator parameters during adversarial training.

Proposed Methodology – Schematic Diagram



Thermal Imaging Datasets Used

Dataset	Images	Key Details
KAIST Multispectral [5]	9,500	Sensor specifications, capture conditions
CBSR NIR Face [6]	3,940	Wavelength range, pose variations
Neonatal Thermal [5]	11,000	Ethical considerations, respiratory syndromes
Sapota Bruise [5]	5,030	Environmental controls, temporal aspects
Chilli Disease [5]	3,000	Growth stages, disease signatures

Datasets selected to evaluate cross-domain robustness under heterogeneous thermal distributions.

Current Progress Status

Completed Work:

- Literature survey and state-of-the-art analysis
- Identification of research gaps
- Problem formulation
- Mathematical modeling of physics constraints

In Progress:

- Implementation of baseline LiquidGAN
- Integration of physics-informed loss

Work Plan for Next Two Months

Month 1:

- Complete baseline LiquidGAN implementation
- Verify training stability and reproduce reported results
- Design and integrate physics-aware loss components
- Implement gradient consistency and energy conservation terms

Month 2:

- Train Physics-Constrained LiquidGAN
- Conduct zero-shot cross-domain evaluation
- Perform hyperparameter tuning (μ_1, μ_2)
- Comparative evaluation with DCGAN and WGAN-GP
- Conduct ablation study

Conclusion

- Identified limitations of existing GAN-based thermal synthesis methods
- Proposed Physics-Constrained LiquidGAN framework
- Integrated thermodynamic constraints into adversarial learning
- Designed zero-shot cross-domain evaluation strategy

Current Status:

- Baseline implementation in progress
- Physics-aware loss integration under development

Expected Contribution:

- Improved thermodynamic consistency
- Better cross-domain robustness under data scarcity

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